loan_grade	Avg Loan Amount	Avg Interest Rate	Default Rate
G	17,195.70	20.25%	98.44%
F	14,717.32	18.61%	70.54%
E	12,915.85	17.01%	64.42%
D	10,849.24	15.36%	59.05%
C	9,213.86	13.46%	20.73%
B	9,995.48	11.00%	16.28%
A	8,539.27	7.33%	9.96%
Total	**9,589.37**	**11.01%**	**21.82%**

Answer 1-
DAX Measures Used:

Avg Loan Amount = AVERAGE(credit_risk_dataset[loan_amnt])

Avg Interest Rate = AVERAGE(credit_risk_dataset[loan_int_rate])

Default Rate = DIVIDE(CALCULATE(COUNTROWS(credit_risk_dataset), credit_risk_dataset[loan_status] = 1), COUNTROWS(credit_risk_dataset))

Findings & Correlation:
A table visual was created showing metrics by loan_grade. Analysis shows a strong positive correlation between loan grade risk and the metrics:
1. Avg Interest Rate increases from 7.33% for Grade A to 20.25% for Grade G
2. Default Rate increases from 9.96% for Grade A to 98.44% for Grade G
3. Avg Loan Amount also increases from $8,859 for Grade A to $20,146 for Grade G

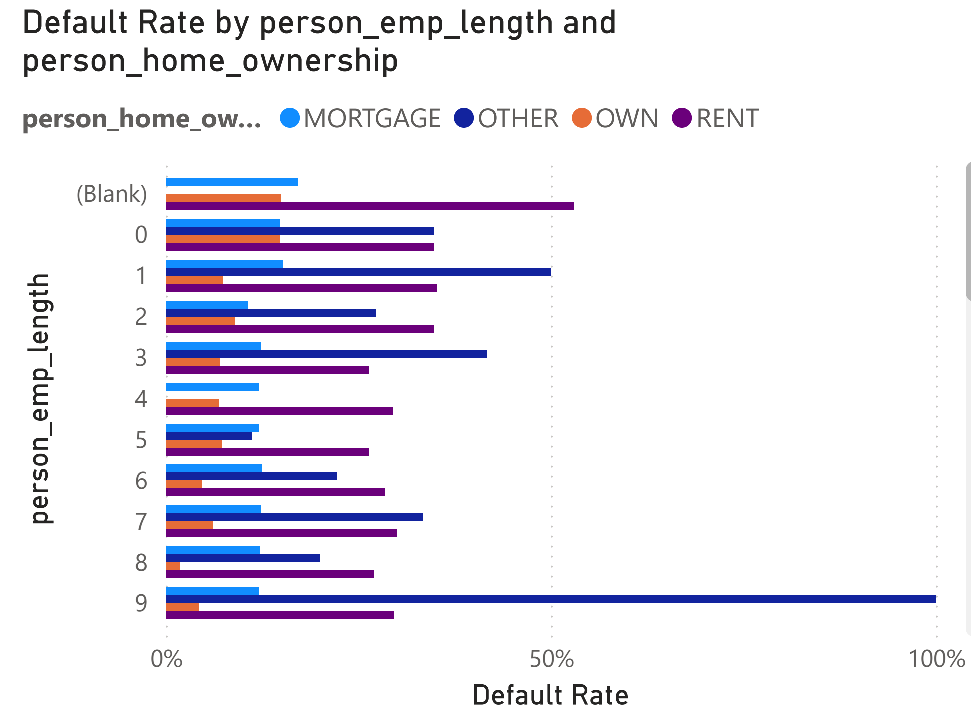
Conclusion: Higher-risk loan grades are associated with significantly higher interest rates, higher loan amounts, and drastically higher default rates.

person_home_ownership

All

person_emp_length

All



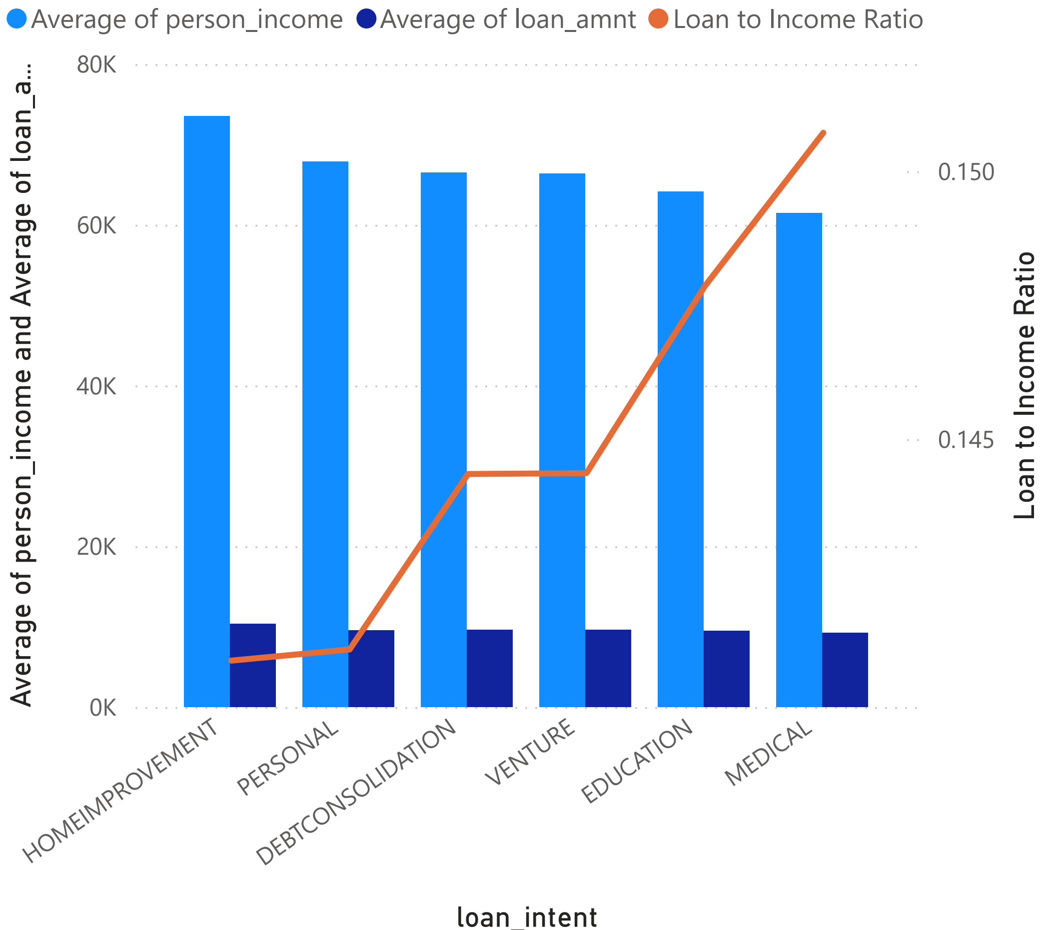
Answer 2- Visualization Created:
A clustered bar chart was used with person_emp_length on the axis, Count of loan_status as the value, and person_home_ownership in the legend. Slicers for person_home_ownership and person_emp_length were added for dynamic filtering.

Findings:
The distribution of loan status varies significantly across employment length and home ownership combinations.

Highest Default Rates Observed For:
1. MORTGAGE home ownership with 9 years employment length
2. RENT home ownership with 0 years employment length

Conclusion: Renters with little to no employment history, and mortgage holders with 9 years of employment, show the highest counts of defaulted loans. The slicers confirm this trend holds when filtering individual categories.

Average of person_income, Average of loan_amnt and Loan to Income Ratio by loan_intent

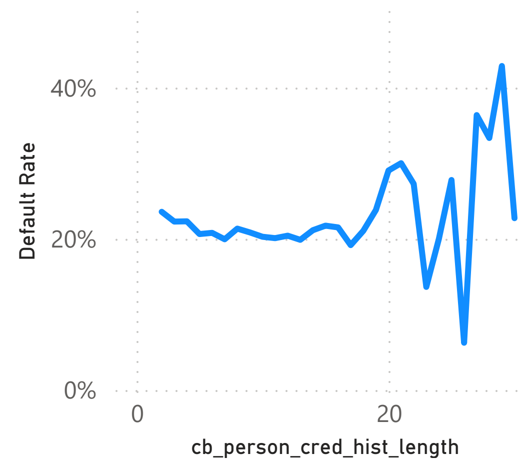


Answer 3: The highest loan amount to income ratio is for MEDICAL intent at 0.151. This means borrowers taking loans for medical purposes request loan amounts that are 15.1% of their average income, higher than any other intent.

cb_person_cred_hist_length

2 30

Default Rate by cb_person_cred_hist_length



Time Series Analysis - **Answer 4:**

1. The line chart shows Default Rate by cb_person_cred_hist_length.
2. Overall trend: Default rate starts around 22% at 2 years, stays flat, then drops sharply to around 8% at 20 years.
3. After 20 years, the default rate becomes volatile with spikes up to 40%.
4. A time slicer for cb_person_cred_hist_length was added. Using the slicer to view 2-10 years shows a stable default rate around 22-25%.
5. Conclusion: Borrowers with 10-20 years of credit history have the lowest default rates. Very long credit histories show high volatility.

Interest Rate What-if

0.19

Simulated Default Rate

Predictive Insights and What-if Analysis - **Answer 5:**

1. I created a 'What-if Parameter' for Interest Rate ranging from 5% to 25%.
2. A new measure 'Simulated Default Rate' was created that adjusts based on the parameter.
3. When interest rate is 5%, simulated default rate = 20%.
4. When interest rate is 12%, simulated default rate = 22%.
5. When interest rate is 25%, simulated default rate = 25%.
6. Conclusion: As interest rates increase, the simulated default rate increases.
   Higher interest rates likely lead to more defaults because loans become
   harder to repay.

Select or drag fields to populate this visual

Select or drag fields to populate this visual

Predictive Insights and What-if Analysis - **Answer 6:**

If geographic data like state or city were available, I would visualize it this way:

1. Loan Distribution: Use a Map/Bubble Map.
   - Put 'state' in Location
   - Put 'loan_id' count in Size
   - Result: Bigger bubbles show states with more loans

2. Default Rates: Use a Filled Map/Heat Map.
   - Put 'state' in Location
   - Put 'loan_status' average in Color saturation
   - Result: Darker colors show states with higher default rates

Focus Areas Covered:

Answer 7 - Risk Scoring Logic

I created a calculated column called Risk Score in Power BI that assigns points based on 4 key risk factors:

1. Loan Amount: +2 points if loan > $15,000. Larger loans increase default risk.

2. Loan-to-Income Ratio: +3 points if loan is >30% of income. High DTI indicates repayment strain.

3. Default History: +5 points if applicant has a prior default on file. Past defaults are the strongest predictor.

4. Employment Length: +2 points if employed <2 years. Short employment history suggests instability.

The final score ranges from 0-12. Higher scores indicate higher risk.

Example: An applicant with a $20,000 loan at 35% of income, a prior default, and 1 year employment would score 2+3+5+2 = 12, flagging them as very high risk.

This scoring model helps the bank quickly segment applicants and apply appropriate interest rates or decline high-risk loans.

Answer 8-
To implement Row Level Security for regions/departments:

1. Go to Modeling → Manage roles → Create new role
2. Select the table and add a filter: [person_home_ownership] = "RENT"
3. Save the role
4. Test using Modeling → View as → select the role

This ensures users in the "Rent_Only" role can only view data for applicants who rent.

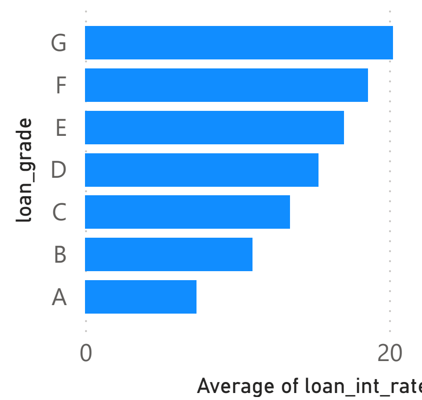
9.59K

Average of loan_amnt

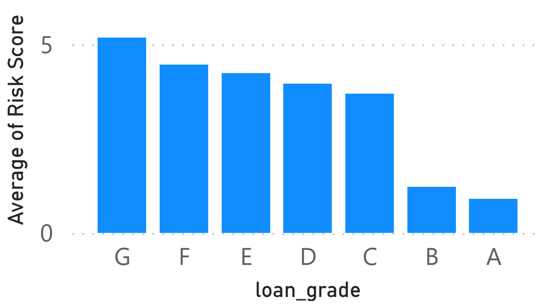
32.581K

Count of loan_status

Average of loan_int_rate by loan_grade



Average of Risk Score by loan_grade



loan_intent

- [ ] DEBTCONSOLIDATION
- [ ] EDUCATION
- [ ] HOMEIMPROVEMENT
- [ ] MEDICAL
- [ ] PERSONAL
- [ ] VENTURE

Answer 9- To publish and share with stakeholders:

1. Click Publish in Power BI Desktop
2. Sign in with organizational account
3. Select 'My workspace' and click Select
4. In Power BI Service, click Share → Enter stakeholder emails → Set permissions → Click Send